

MADHUHRITA SAHA

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ECE undergraduate (4th year, GPA 9.04) with three years of continuous, hands-on engineering design practice spanning VLSI circuit simulation, FPGA prototyping, IoT system design, and ML-integrated hardware projects. Proficient in Cadence Virtuoso, Xilinx Vivado, and Python for simulation, analysis, and validation. Published researcher (2 papers) with a strong record of designing complex & ML-driven engineering problems from concept through hardware realization. Also, a creative enthusiast passionate about AIoT and embedded systems for sustainable smart infrastructure. Experienced at documenting end-to-end design flows, edge cases, and validation criteria. Equipped with strong technical foundations, leadership abilities, presentation, problem-solving and creative design thinking skills.

TECHNICAL SKILLS & SIMULATION TOOLS

Simulation & Design Tools: Cadence Virtuoso (CMOS transistor-level), Xilinx Vivado (FPGA synthesis, P&R, timing), Cisco Packet Tracer (network simulation), MATLAB

HDL & Scripting: Verilog HDL (behavioral/dataflow/structural), Python (TensorFlow, Keras, NumPy, Pandas, Scikit-learn), C/C++

Engineering Domains: VLSI & Digital Design, Embedded Systems (ESP32, Arduino), IoT, Edge AI, MOS Gas Sensors / E-Nose, Applied ML/CV, Circuit Theory

Design Skills: RTL coding, CMOS layout, FPGA prototyping, combinational/sequential circuit design, physical layout, sensor fusion & feature engineering

Documentation & Comms: Research paper writing, technical presentations, academic publication, LaTeX

ENGINEERING DESIGN PROJECTS

The following projects span 3 years of continuous engineering design, spanning electrical, embedded, IoT, and AI domains. Each involved trade-off analysis, simulation or hardware validation, and complete documentation.

Electronic Nose-Based VOC Detection & Edge AI System (IIT Mandi)

2026

- Designed and developed a low-cost, standalone Electronic Nose (E-Nose) integrating a 5-sensor MOS array and custom acrylic sensing chambers for VOC detection.
- Programmed ESP32 Embedded C/C++ firmware for real-time multi-sensor acquisition, statistical feature extraction, and on-device ML inference with a Wi-Fi dashboard.
- Evaluated PCA, Random Forest, and Decision Trees achieving 65% cross-validated accuracy while identifying and mitigating environmental confounding factors.

Bridge Health Monitoring System

2026

- Designed a sensor-fusion system for real-time structural integrity monitoring of bridges, integrating ML model, accelerometers, and vibration sensors with microcontroller-based data acquisition.
- Performed signal conditioning circuit design, threshold analysis, and alert logic for anomaly detection — documented complete design trade-offs between power consumption and sampling resolution.

Analog Voice Modulator

2026

- Designed and implemented an analog signal processing pipeline for voice modulation, covering filter design (bandpass, notch), amplifier biasing, and frequency shifting.
- Analyzed distortion, signal-to-noise trade-offs, and component tolerances; validated output waveforms against design specifications.

AI-Powered IoT Microclimate Control System for Smart Indoor Farming

2025 — Ongoing

- Architected an RL-based edge intelligence system on ESP32 for real-time optimization of temperature, humidity, CO2, light, and irrigation — requiring multi-parameter trade-off optimization.
- Integrated XAI (Explainable AI) to generate human-interpretable decision rationale for each control action; designed sensor calibration and validation pipelines.
- Documented boundary conditions, failure modes, and system edge cases for real-world deployment validation.

Alcohol Detection & Engine Locking System with Drowsiness Detection

2025

- Designed a hybrid vehicle safety prototype combining breath-alcohol sensor interfacing with ML/CV-based eye-movement drowsiness detection; integrated hardware and software into a unified safety decision loop.
- Defined threshold parameters, false-positive/negative edge cases, and fail-safe logic — translating real-world safety constraints into verified system behavior.

Toxic Gas Detection & Filtration Activation System for Firefighters

2024

- Developed a two-stage system: gas sensor array for multi-gas level detection and automatic filtration activation trigger for firefighter protective gear.
- Designed sensor selection trade-offs (sensitivity vs. specificity), calibration curves, and response-time constraints under high-risk deployment conditions.

Smart Dustbin — Automated Waste Management

2024

- Prototyped a smart waste detection system with ultrasonic fill-level sensing, GPS tracking module integration, and automated municipality alert logic.
- Designed power management trade-offs for battery-operated outdoor deployment and documented communication protocol selection rationale.

Smart Kitchen Cabinet

2023

- Led a team to design a home automation cabinet with load cell-based weight sensing per compartment, ESP32-CAM image recognition for contents, and integrated door display.
- Managed system integration, sensor calibration, and end-to-end hardware-software co-design documentation.

INTERNSHIPS

Summer Research Intern

May 2026 – Jul 2026

Indian Institute of Technology, Mandi · Mandi, Himachal Pradesh

- Designed and developed a low-cost, standalone Electronic Nose (E-Nose) system for non-destructive onion freshness classification using a 5-sensor MOS array and custom-fabricated acrylic sensing chambers.
- Developed Embedded C/C++ firmware on ESP32 for real-time VOC sensor data acquisition, statistical feature extraction, and on-device ML inference, integrated with a Wi-Fi web dashboard for monitoring and logging.
- Engineered statistical features from multi-sensor data and evaluated PCA, Random Forest, and Decision Tree classifiers, achieving 65% cross-validated accuracy on the deployable sensor-only edge model.
- Identified and mitigated environmental confounding factors in sensor data, successfully demonstrating a robust end-to-end edge AI pipeline from VOC sensing to on-device freshness prediction.

Microelectronics Technology & VLSI Design Intern | Grade: A

Dec 2025 – Feb 2026

Jadavpur University, Kolkata

- Designed and simulated CMOS transistor-level circuits (inverters, NAND, NOR, AND, XOR) in Cadence Virtuoso, analyzing propagation delay, power dissipation, and noise margin trade-offs.
- Performed FPGA-based prototyping in Xilinx Vivado covering simulation, synthesis, place & route, timing analysis, and hardware verification — complete HDL-to-hardware design flow.
- Implemented and validated combinational and sequential circuits: adders, multiplexers, decoders, counters, LCD interfacing, and 7-segment display controllers using Verilog HDL.
- Gained IC physical layout design exposure; documented design decisions, edge cases, and design rule check (DRC) outcomes for each implemented circuit.

AI & IBM Cloud Technologies Intern

Jul 2025 – Aug 2025

Edunet Foundation in collaboration with AICTE

- Designed and deployed a healthcare-focused AI chatbot on IBM Cloud, integrating AutoAI for model selection and data analytics for response validation.

AICTE Edunet–Shell Intern — Green Skills Programme

Jun 2025 – Jul 2025

Shell India Markets Pvt. Ltd. & Edunet Foundation, AICTE

- Built and deployed an e-waste image classifier using EfficientNetV2B0 (Python, TensorFlow, Keras, Gradio); documented model performance trade-offs and validation on held-out test sets.

Virtual Intern — Cybersecurity & AI

Jul 2025

NIIT Foundation

- Executed network simulation and validation projects using Cisco Packet Tracer; documented topology design decisions and boundary conditions.

PUBLICATIONS & RESEARCH

- Saha, M.; Roy, M.; Dey, K. Innovations in Blood Pressure Measurement: A Review of Techniques and Future Trends. SD 2026, 2(2), 65–70. <https://doi.org/10.54162/SD02-260215/05>
- Madhuhrita Saha et al. Unlocking Insights: The Power of Big Data Analytics. American Journal of Electronics & Communication, Vol. V(1), 106–112.
- The Convergence of AI and IoT: A Review of AIoT. Presented at ICSAA-2025 (ISBN 978-93-341-6437-4).
- AI-Enabled Energy Forecasting using Enhanced LSTM and XGBoost Hybrid Model. Scheduled for inclusion in OPTO VLSI and AI Driven Remote Sensing (upcoming book chapter).

EDUCATION

B.Tech in Electronics & Communication Engineering

2023 – 2027 (Ongoing)

University of Engineering and Management, Kolkata (UEMK) | Avg GPA: 9.04

Relevant Coursework: Digital Electronics & VLSI, Embedded Systems, AI/ML, Signals & Systems, Analog Circuits, Microprocessors

Higher Secondary (PCMB)

Holy Child Institute, Kolkata | WBCHSE | 84.2%

2023

Secondary

Holy Child Institute, Kolkata | WBBSE | 91%

2021

CERTIFICATIONS & ACHIEVEMENTS

- VLSI SoC Design using Verilog & HDL — Maven Silicon
- AI for Organizational Leaders — Microsoft & LinkedIn Learning
- Internet of Things: How did we get here? — Coursera, UC San Diego
- Cybersecurity Fundamentals — Coursera, University of London
- Accenture Nordics Software Engineering Job Simulation — Forage
- Student Excellence Award 2025 (Academic Year 2024–25)
- Presented research & review papers at UEMCOS 2024, ICSAA-2025, iC2IS-2025, UEMCOS 2025, UEMGREEN 2026
- 14-week Entrepreneurship Programme — Wadhwani Foundation, NEN

REFERENCES

- Dr. Subhajit Das — Associate Professor, IEM UEM Kolkata | subhajit.das@uem.edu.in | +91 9674639245
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